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The yield and ultimate tensile strength of steel welds

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Abstract

The design of welding alloys to match the ever advancing properties of newly developed steels is not an easy task. It is traditionally attained by experimental trial and error, modifying compositions and welding conditions until a satisfactory result is discovered. Savings in cost and time might be achieved if the trial process could be minimised. This work outlines the use of an artificial neural network to model the yield and ultimate tensile strengths of weld deposits from their chemical composition, welding conditions and heat treatments. The development of the models is described, as is the confirmation of their metallurgical accuracy. © 1997 Elsevier Science S.A.

Keywords: Neural network; Strength; Welding alloys

1. Introduction

The tensile test provides the basic design data essential in both the specification and acceptance of welding materials. The test itself is simple but nevertheless gives reproducibly the parameters describing the onset of plastic deformation (the yield strength), an estimate of the maximum load that can ever be supported (the ultimate tensile strength) and a variety of measures of ductility. Although the measurements involved are straightforward, their values depend in a complicated way on the chemical composition, the welding parameters and any heat treatment. There is, therefore, no general fundamental or empirical model capable of estimating the tensile parameters as a function of all these variables [1,2]. This is in spite of the fact that a great deal is known about the mechanisms of strengthening.

The difficulty limiting progress is the complexity of the problem. The physical models for strengthening mechanisms are not sufficiently sophisticated [3] and the linear regression methods used traditionally [4–20] are not capable of representing the real behaviour which is far from linear when all the factors are taken into account.

The aim of this work was to use an artificial neural network to empirically model and interpret the dependence of the yield and ultimate tensile strength of steel weld deposits as a function of many variables. A neural network is capable of realising a great variety of nonlinear relationships of considerable complexity. Data are presented to the network in the form of input and output parameters, and the optimum non-linear relationship found by minimising a penalised likelihood. As in regression analysis, the results then consist of a series of coefficients (called weights) and a specification of the kind of function which in combination with the weights relates the inputs to the outputs. To prevent overfitting, MacKay [21–24] has developed a Bayesian framework to control a neural network's complexity. This framework also provides quantified error bars on the network's predictions and makes it possible to automatically identify which of many possibly-relevant input variables are in fact important factors in the regression.

The creation of a model using the neural network method requires data and thus a large database of experimental measurements was assembled for analysis.

2. The experimental database

The definition of strength in any model ideally requires a complete description of the chemical composition, the welding parameters, the heat-treatment and various features of the microstructure. A search of the

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literature [4–20,25–54] showed that most of the reported data are not as detailed as these requirements. Given that the neural network technique used cannot cope with missing values, it was necessary to limit the number of variables to take maximum advantage of the 1652 individual cases studied.

Table 1
The variables used in the analysis of (a) the yield strength, (b) factors influencing the ultimate yield strength

Variable	Range	Mean	S.D.
(a)			
Carbon (wt.%)	0.029–0.16	0.074	0.024
Silicon (wt.%)	0.040–1.14	0.34	0.124
Manganese (wt.%)	0.27–2.25	1.20	0.39
Sulphur (wt.%)	0.001–0.14	0.0097	0.0069
Phosphorus (wt.%)	0.004–0.25	0.013	0.011
Nickel (wt.%)	0.00–3.50	0.22	0.63
Chromium (wt.%)	0.00–9.35	0.734	2.07
Molybdenum (wt.%)	0.00–1.50	0.17	0.35
Vanadium (wt.%)	0.00–0.24	0.018	0.049
Copper (wt.%)	0.00–1.63	0.074	0.224
Cobalt (wt.%)	0.00–2.80	0.011	0.147
Tungsten (wt.%)	0.00–2.99	0.0115	0.146
Titanium (ppm wt.)	0–690	40.86	79.9
Boron (ppm wt.)	0–69	1.17	5.78
Nobium (ppm wt.)	0–1000	57.4	1.51
Heat input (kJ mm^{-1})	0.6–7.9	1.85	1.47
Interpass temperature (°C)	20–300	207.7	48.93
Tempering temperature (°C)	0–760	320.4	257
Tempering time (h)	0–24	5.72	6.29
Yield strength (MPa)	315–920	507.3	92.8
(b)			
Carbon (wt.%)	0.029–0.160	0.076	0.018
Silicon (wt.%)	0.04–1.14	0.35	0.14
Manganese (wt.%)	0.27–2.22	1.23	0.36
Sulphur (wt.%)	0.004–0.14	0.010	0.007
Phosphorus (wt.%)	0.004–0.25	0.012	0.012
Nickel (wt.%)	0.00–3.15	0.16	0.53
Chromium (wt.%)	0.00–9.35	0.62	1.91
Molybdenum (wt.%)	0.00–1.50	0.18	0.36
Vanadium (wt.%)	0.00–0.24	0.02	0.04
Copper (wt.%)	0.00–1.63	0.07	0.23
Cobalt (wt.%)	0.00–2.80	0.02	0.18
Tungsten (wt.%)	0.00–2.99	0.02	0.18
Oxygen (ppm wt.)	132–1650	441	152
Titanium (ppm wt.)	0–690	49	91
Boron (ppm wt.)	0–69	1.15	6.18
Nobium (ppm wt.)	0–985	62	165
Heat input (kJ mm^{-1})	1–7.9	1.9	1.5
Interpass temperature (°C)	100–300	205	51
Tempering temperature (°C)	0–760	317	243
Tempering time (h)	0–16	6.1	6.2
Yield strength (MPa)	329–920	503	88
Ultimate tensile strength (MPa)	447–1151	590	92

ppm wt., parts per million by weight.

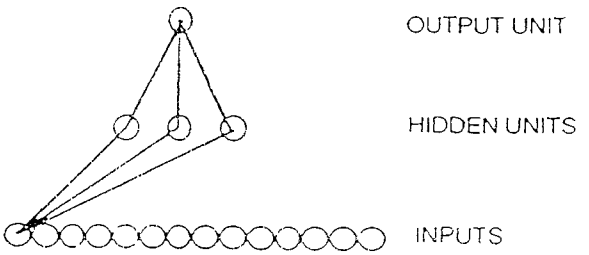


Fig. 1. A typical network used in the analysis. Only the connections originating from one input unit are illustrated and the two bias units are not illustrated.

Table 1(a) shows the list of 19 input variables used for the yield strength analysis. It was possible to find 770 cases where all of these variables have been reported, with the exception of phosphorus and sulphur concentrations. It would have been unreasonable to set these latter concentrations to zero when their values are not reported; welds inevitably contain these impurity elements in practice. For cases where the P and S values were missing, their concentrations were set to the mean values calculated for the 1652 case database. For the other elements such as Mn, Ni, etc., the concentrations can truly be set to zero when they are not reported. This is because they would not then be deliberate additions and would have concentrations close to the limits of chemical analysis with the techniques generally used.

The same rationale was applied to the dataset for the ultimate tensile strength analysis (Table 1(b)). Unlike the yield strength, the UTS can be sensitive to the inclusion content since there are much larger plastic strains associated with the UTS. Failure mechanisms such as void formation (at inclusions) may then contribute to the formation of a maximum in the load-extension curve. For this reason, the oxygen concentration was included as an additional variable. It is well-established that the UTS correlates strongly with the yield strength [55] and in addition, the UTS must always be larger then the yield strength. It was found that the latter condition was best satisfied during modelling by including the yield strength as a variable in the UTS analysis. Because of the inclusion of oxygen as a variable, the UTS dataset consisted of fewer cases (520 in all), each with 21 input variables. Of these 520 cases, 18 did not report the yield strength values, this was resolved by substituting values calculated using the yield strength model.

3. Brief description of neural network

The aim is to be able to estimate the yield strength or UTS as a function of the variables listed in Table 1.

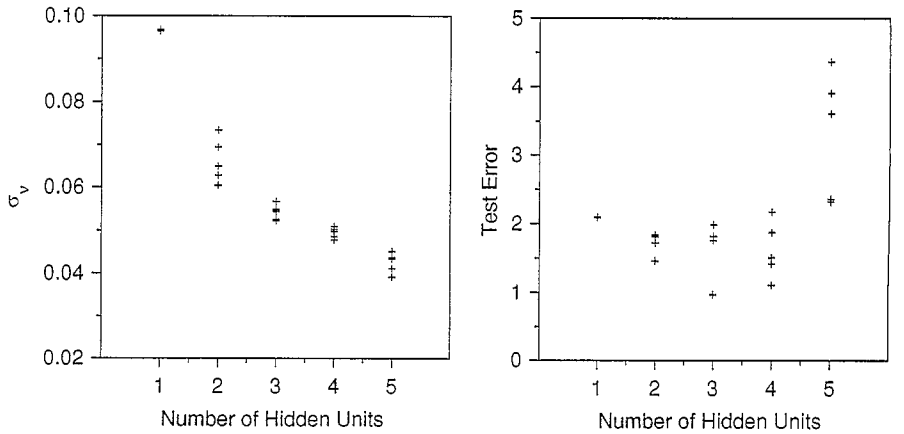


Fig. 2. Variation in σ_v and in test error as a function of the number of hidden units. Several values are presented for each set of hidden units because the training for each network was started with a variety of random seeds.

Table 2
Ranking of ten best models by test error

Ranking	Hidden units	Seed	Test energy	Ranking	Hidden units	Seed	Test energy
1	3	35	0.96311	6	2	100	1.4566
2	3	50	0.96994	7	4	100	1.5045
3	4	35	1.1063	8	2	15	1.7185
4	4	5	1.4146	9	3	100	1.755
5	2	50	1.4556	10	2	35	1.8094

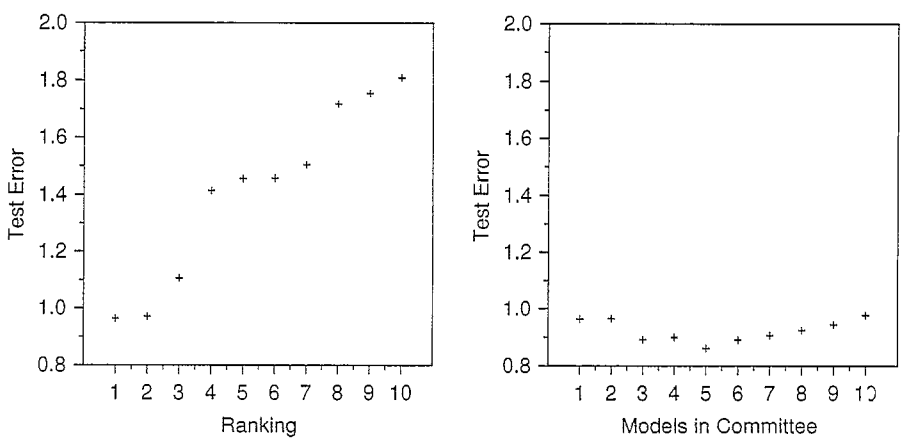


Fig. 3. Test error of the top ten ranked yield strength models and test error of the committees of yield strength models.

Both the input and output variables were first normalised within the range ± 0.5 as follows:

$$x_N = \frac{x - x_{min}}{x_{max} - x_{min}} - 0.5 \quad (1)$$

where x_N is the normalised value of x which has maximum and minimum values given by x_{max} and x_{min} respectively. This normalisation is not necessary for the analysis, but enables an easier subsequent comparison of the significance of each of the variables.

Several neural network models were created using the data; the models varied in terms of their complexity and in terms of the random seeds used to start the training process.

Each network consisted of input nodes (one for each variable x), a number of hidden nodes and an output node (representing either the yield or ultimate tensile strength, Fig. 1). Linear functions of the inputs x_j are operated on by a hyperbolic tangent transfer function:

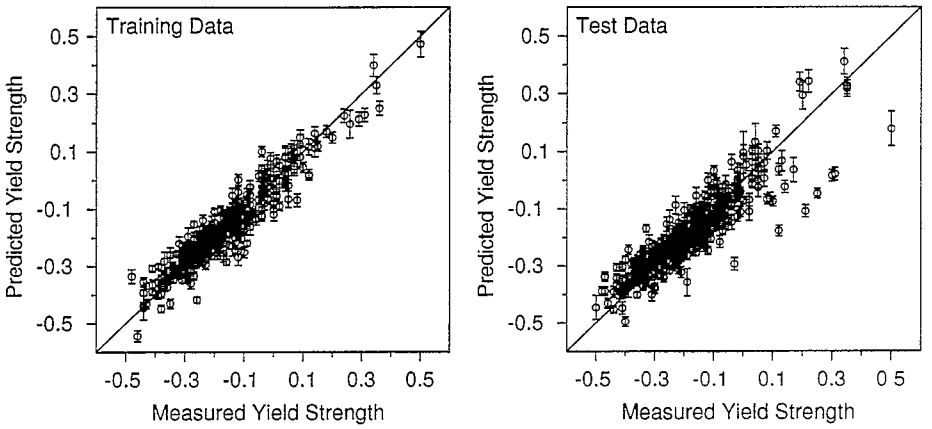


Fig. 4. Training and test data for the single highest ranked model.

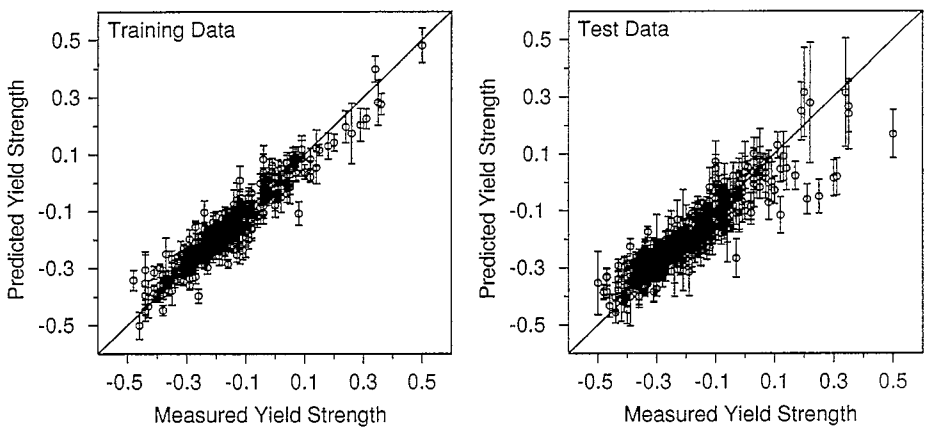


Fig. 5. Training and test data for the committee consisting of the five highest ranking models.

$$h_i = \tanh\left(\sum_j w_{ij}^{(1)} x_j + \theta_i^{(1)}\right) \quad (2)$$

so that each input contributes to every hidden unit. The bias is designated θ_i and is analogous to the constant that appears in linear regression. The strength of the transfer function is in each case determined by the weight w_{ij}. The transfer to the output y is linear:

$$y = \sum_i w_{ij}^{(2)} h_i + \theta^{(2)} \quad (3)$$

The specification of the network structure, together with the set of weights is a complete description of the formula relating the inputs to the output. The weights are determined by training the network [21–24]. The training involves a minimisation of the regularised sum of squared errors. The objective function which is minimised is described in [21–23]. The term σ_v used below is the framework estimate of the noise level of the data.

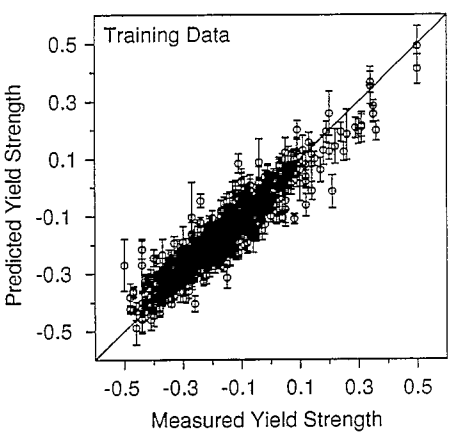


Fig. 6. Training data for the committee produced from the five highest ranking models, trained on the whole data set.

4. The yield strength model

The method was applied to the yield strength data (Table 1(a)). The complexity of the model is controlled by the number of hidden units (Fig. 1), and the values of the regularisation constants (σ_w), one associated with each of the inputs, one for biases and one for all the

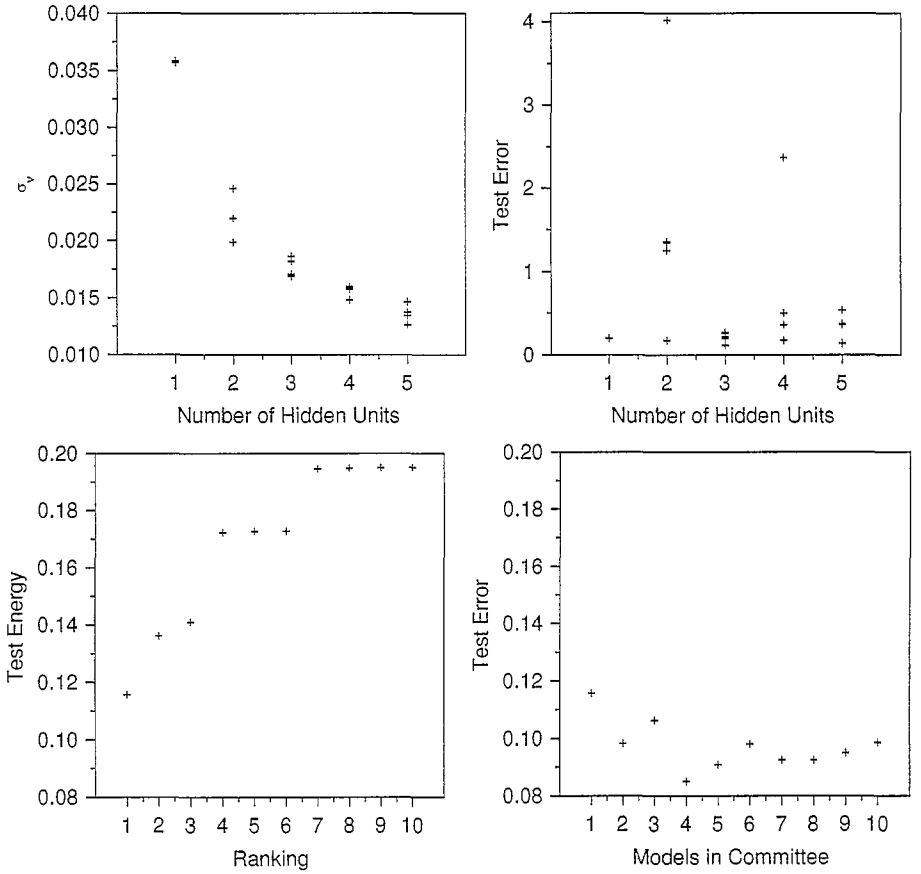


Fig. 7. (a) Variations in σ_ν and in the test error as a function of the number of hidden units. Several values are presented for each set of hidden units because the training for each network was started with a variety of random seeds. (b) Test error of the top ten ranked UTS models and the test error of the committees of UTS models.

weights connected to the output. Fig. 2 shows that as expected, the inferred noise level decreases monotonically as the number of hidden units increases. However, the complexity of the model also increases with the number of hidden units. A high degree of complexity may not be justified if the model attempts to fit the noise in the experimental data. In addition, the experimental data were partitioned equally and randomly into a test dataset and a training dataset. Only the latter was used to train the model, whose ability to generalise was examined by checking its performance on the unseen test data. The test error (T_{en}) is a reflection of the ability of the model to predict the yield strength values in the test data:

$$T_{en} = 0.5 \sum_n (y_n - t_n)^2 \quad (4)$$

where y_n is the set of predictions made by the model and t_n is the set of target (experimental) values.

It can be seen that the calculated test error for this yield strength model goes through a minimum at 3 hidden units (Fig. 2(b)). This would be the optimum model if a single model were to be used for the analysis.

However, it is possible that a *committee* of models can make a more reliable prediction than an individual model. The best models were ranked using the values of their test energies (Table 2 and Fig. 3). Committees could then be formed by combining the prediction of the best L models, where $L = 1,2,\cdots$; the size of the committee is therefore given by the value of L.

The test error of the predictions made by a committee of L models, ranked $1,2\cdots l\cdots L$, each with n lines of test data, is calculated in a similar manner to the test error of a single model.

$$T_{en} = 0.5 \sum_n (\overline{y_n} - t_n)^2 \quad \text{where} \quad \overline{y_n} = \frac{1}{L} \sum_l y_n^{(l)} \quad (5)$$

The test energies for the first ten committees (with ever increasing size) are plotted in Fig. 3. The two plots in Fig. 3 are drawn to the same scale to emphasise the reduction in test error and hence improved predictions, by using the committee model approach.

It is evident that the test error for the committee analysis goes through a minimum for the committee made up of five models. These models were used in calculating the optimum result. The comparison be-

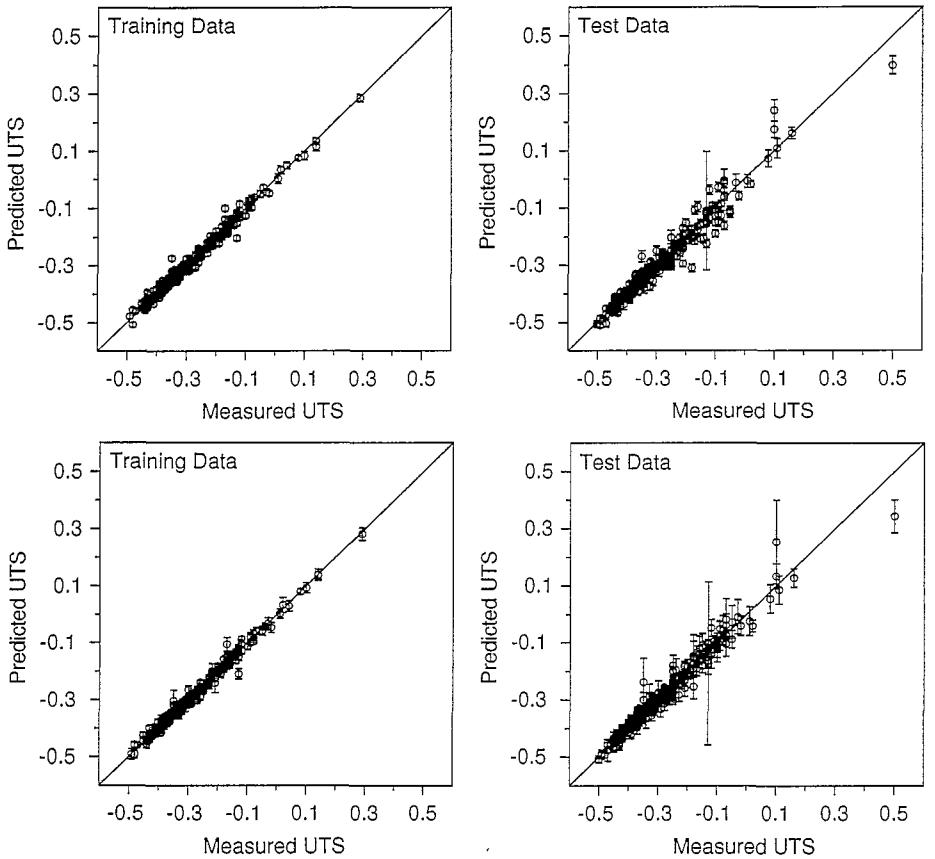


Fig. 8. (a) Training and test data for (a) the highest ranked model, (b) the committee produced from the four highest ranking models.

tween the predicted and measured values of the yield strength for the training and test data are shown for the best single model and the best committee (consisting of five best models) in Figs. 4 and 5. For this purpose, the committee prediction is given by the mean value of its components:

$$\bar{y} = \frac{1}{L}\sum_l y^{(l)} \tag{6}$$

and the committee variance is given by [56]:

$$\sigma^2 = \frac{1}{L}\sum_l \sigma_y^{(l)2} + \frac{1}{L}\sum_l (y^{(l)} - \bar{y})^2 \tag{7}$$

It is notable that the committee is more realistic than the single model in the sense that there are very few outliers i.e., for the test data there are few points for which the predictions are out of the 1σ error bar range.

Having chosen the best committee, its five component models were retrained on the entire dataset, the retraining procedure beginning with the weights already determined. The results from this ultimate committee model are shown in Fig. 6. It is satisfying that outliers are virtually absent.

5. Model for ultimate tensile strength

The technique discussed above was then applied to the UTS dataset (Table 1(b)). The plots of σ_y, test energies and the appropriate plots for the committee investigation are shown in Fig. 7. The estimates for the training and test data for the best single model and the best committee (consisting of four of the best models) are shown in Fig. 8. Finally, the four models making up the committee were retrained on all data, giving the final committee model as shown below in Fig. 9.

6. Application of the models

In this section we examine the influence of each of the variables on the yield and tensile strengths of three welding alloys which are the subject of a separate experimental programme. Their actual compositions and mechanical properties are listed in Table 3. The welds, which are fabricated by Oerlikon, cover a range of applications in power plant construction, being referred to in brief as $2\frac{1}{4}$Cr1Mo, 9Cr and $3\frac{1}{2}$N deposits. The purpose of the analysis presented below was to test the neural network models to see whether the relation-

ships recognised are as expected from metallurgical considerations.

Committee models are used for all the predictions that follow; the calculation of the error bar for the committee, from the corresponding errors for each model within the committee, has already been discussed. However, the UTS model requires the yield strength for one of its inputs, which in turn is calculated using the yield neural network committee. Errors in the calculation of yield strength must therefore add to the estimate of UTS:

$$\sigma^2 = \frac{1}{L}\sum_l \sigma_y^{(l)2} + \frac{1}{L}\sum_l (y^{(l)} - \bar{y})^2 + \sigma^{(y)2}\left(\frac{\partial u}{\partial y}\right)^2 \quad (8)$$

where the notation is as before, with the exception of the final term which represents the error in the yield calculation multiplied by the sensitivity of the UTS to yield.

The calculated yield and UTS values of the three experimental weld deposits are compared to the measured values in Table 4. These three welds have been included in the training data. There are two calculated values of the UTS: one using the measured yield strength and the other a calculated yield value as an input in the UTS model. As might be expected, the latter method has larger uncertainty associated with it. It is evident that the predictions are all rather good, so further calculations were carried out to check the effects of each of the inputs.

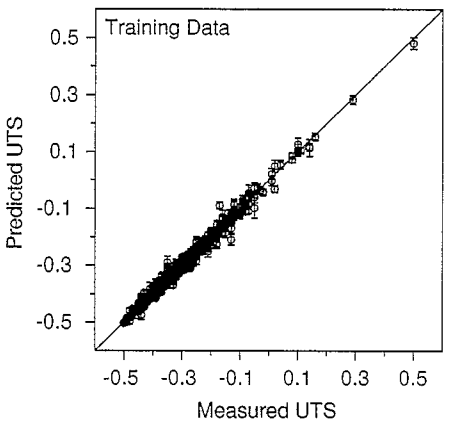


Fig. 9. Training data for the committee produced from the four highest ranking models, trained on the *whole* data set.

Table 3
Chemical composition, welding process and heat treatment data for the three test welds

Variable	$2\frac{1}{4}$Cr1Mo	9Cr	$3\frac{1}{2}$Ni
Carbon (wt.%)	0.07	0.10	0 13
Silicon (wt.%)	0.39	0.38	0.37
Manganese (wt.%)	0.90	1.14	1.47
Sulphur (wt.%)	0.007	0.007	0.007
Phosphorus (wt.%)	0.010	0.008	0.010
Nickel (wt.%)	0.02	0.00	2.97
Chromium (wt.%)	2.64	9.00	1.07
Molybdenum (wt.%0	1.18	1.14	0.66
Vanadium (wt.%)	0.02	0.22	0.049
Copper (wt.%)	0.00	0.00	0.00
Cobalt (wt.%)	0.00	0.00	0.00
Tungsten (wt.%)	0.00	0.00	0.00
Oxygen (ppm wt.)	410	460	200
Titanium (ppm wt.)	0	0	0
Boron (ppm wt.)	0	0	0
Niobium (ppm wt.)	60	600	50
Nitrogen (ppm wt.)	60	460	60
Heat input (kJ mm^{-1})	1.605	1.12	1.20
Interpass temperature (°C)	300	300	300
Tempering temperature (°C)	700	760	700
Tempering time (h)	3	2	3
Yield strength (MPa)	568	752	920
UTS (MPa)	676	872	1000

7. Major alloying elements

The $2\frac{1}{4}$Cr1Mo welds have a bainitic microstructure on deposition, whereas the 9Cr and $3\frac{1}{2}$Ni are martensitic on deposition; the effect of the tempering is to precipitate carbides. Consequently, carbon is in these circumstances expected to raise the strength, although the rise in strength should not be very large since the welds are all severely tempered. This general behaviour is exactly what the model predicts, as illustrated in Fig. 10. The error bars in all cases get larger as the carbon concentration increases beyond the levels typically employed in welds (< 0.16 wt.%), since such data do not exist in the training dataset. It is well known that the ratio of the yield strength to the tensile strength decreases as the former increases; this is reproduced correctly by the model (Fig. 10).

The model also indicates that the yield strength is virtually a linear function of the carbon concentration, as assumed in most linear regression models [3–20]. In fact, the gradients of all the yield strength plots in Fig. 10 are very similar so that the linear relationship is valid across the alloys. This is illustrated in Fig. 11 where the difference between the predicted yield strength for a specific carbon concentration and the predicted yield strength at zero carbon (Δ yield strength) is plotted against the carbon concentration. These slopes are effectively identical to each other for the three steels within the error limits of the analysis.

The predicted effect of silicon is shown in Fig. 12; the calculations for large silicon concentrations are too uncertain to be useful. However, for concentrations less than about 0.5 wt.% the silicon clearly raises the yield strength, by an amount which within the error bounds is consistent with solid solution strengthening at about

Table 4
Measured and predicted yield strength and UTS values for the three steels described in Table 3 and used as a basis for the predictions

Steel	Measured yield strength (MPa)	Predicted yield strength (MPa)	Measured UTS (MPa)	Predicted UTS (MPa) (measured YS)	Predicted UTS (MPa) (calculated YS)
$2\frac{1}{4}$Cr	568	561 ± 29	676	685 ± 8	685 ± 21
9Cr	752	704 ± 36	872	875 ± 13	838 ± 27
$3\frac{1}{2}$Ni	920	915 ± 43	1000	998 ± 10	994 ± 32

The error limits represent 1σ ($\pm 65\%$).

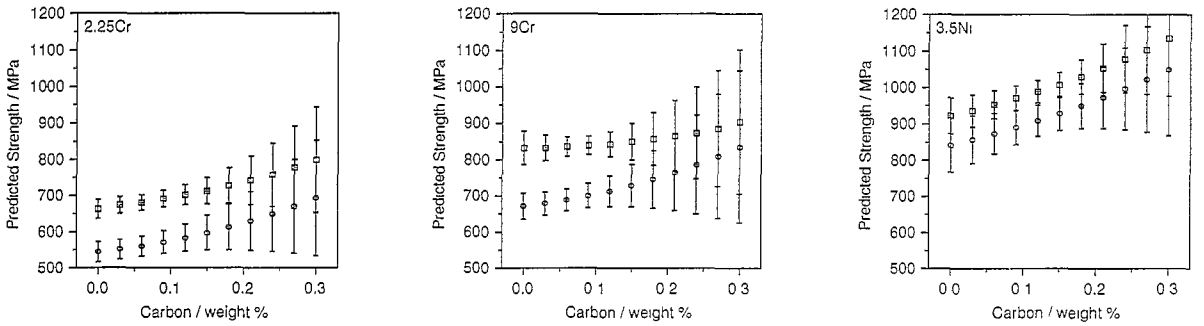


Fig. 10. Predicted variations in yield strength and UTS with carbon variation.

105 MPa/wt.% [57,58]. Silicon is also known to retard the formation of cementite (Fe_3C) [59], but the severe tempering heat treatments used here take the steels well beyond cementite into the secondary hardening regime so that this particular effect must be negligible. Silicon does decrease the lattice parameter of the ferrite matrix which increases the intensity of secondary hardening [60], but the severe tempering again rules this out as an important factor since the carbide particles are expected to be beyond the stage of coherency. It is therefore concluded that the major effect of silicon in these welds is via solid solution strengthening, consistent also with the fact that it does not precipitate or partition into the carbides.

Manganese has a much smaller solid solution strengthening effect (45 MPa/wt.%) [61], which is consistent with the results for the $2\frac{1}{4}$Cr1Mo and $3\frac{1}{2}$Ni welds (Fig. 13). However, the yield strength of the 9Cr weld seems insensitive to the manganese concentration. It is speculated that this may be due to the large overall solute content of this alloy.

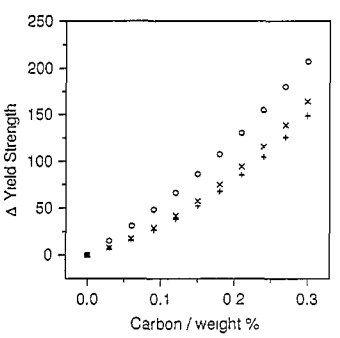


Fig. 11. Plot showing the predicted yield strengh minus the predicted yield strength for zero carbon for the three alloys. + represents the $2\frac{1}{4}$Cr weld, × the 9Cr weld and ○ the $3\frac{1}{2}$Ni weld.

Nickel has an even smaller solid solution strengthening effect than manganese and yet it appears to greatly strengthen all of the welds (Fig. 14). A recent experimental study [62] has demonstrated that austenite can unintentionally form in welds which are post-weld heat treated in cases where the nickel concentration is large.

Nickel is an austenite stabiliser. A phase diagram calculation package, MTDATA [63], was used to predict the phases present at the tempering temperatures for the three welds, for a variety of nickel concentrations Fig. 15. It can be seen that as the nickel concentration is increased, the amount of austenite (γ) in the weld at the post weld heat treatment temperature also increases. The important point is that this austenite subsequently decomposes to untempered martensite on cooling to room temperature and hence leads to an abnormal increase in strength [64]. Comparing the two sets of graphs it can be seen that the rate of rise of yield strength and UTS is related to that of austenite fraction in the weld.

8. Carbide-forming and other elements

Fig. 16 shows that the addition of chromium in general causes an increase in strength, though there are detailed differences between the three kinds of welds. The $3\frac{1}{2}$Ni steel is the most sensitive to changes in chromium concentration whereas the yield strength of the 9Cr steel is increased slightly but then levels off at

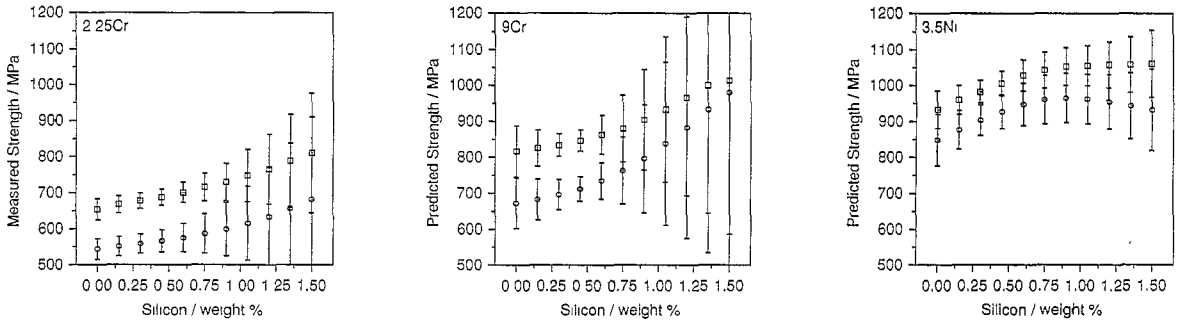


Fig. 12. Predicted variations in yield strength and UTS with silicon variation.

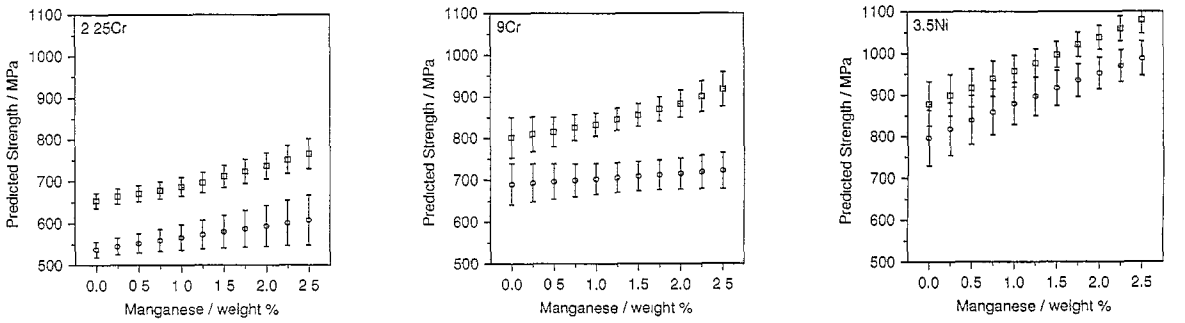


Fig. 13. Predicted variations in yield strength and UTS with manganese variation.

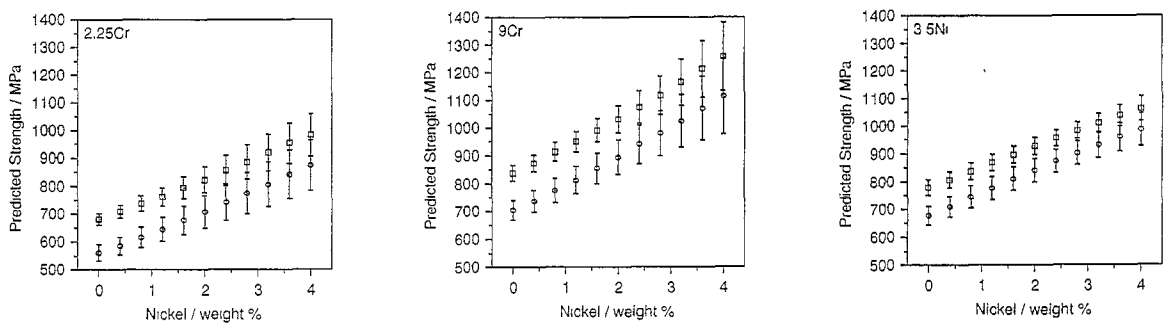


Fig. 14. Predicted variations in yield strength and UTS with nickel variation.

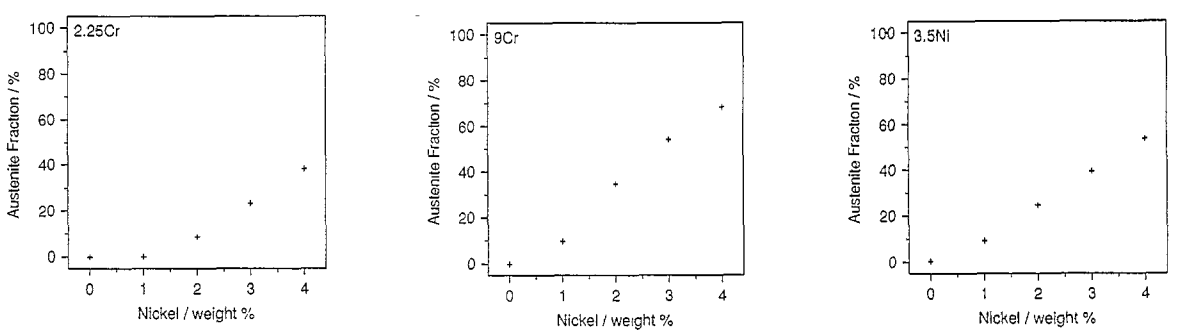


Fig. 15. MTDATA calculation phase present at the tempering temperature for the $2\frac{1}{4}$Cr (700°C), 9Cr (760°C) and $3\frac{1}{2}$Ni (700°C) weld deposits with varying nickel concentration.

about 5 wt.%. The phase stability calculations presented in Table 5 provide some of the explanation for the differences between the alloys. For the 9Cr steel, there is little variation in the total fraction of carbides beyond about 6 wt.% chromium, explaining the predicted lack of increase in strength There is a large

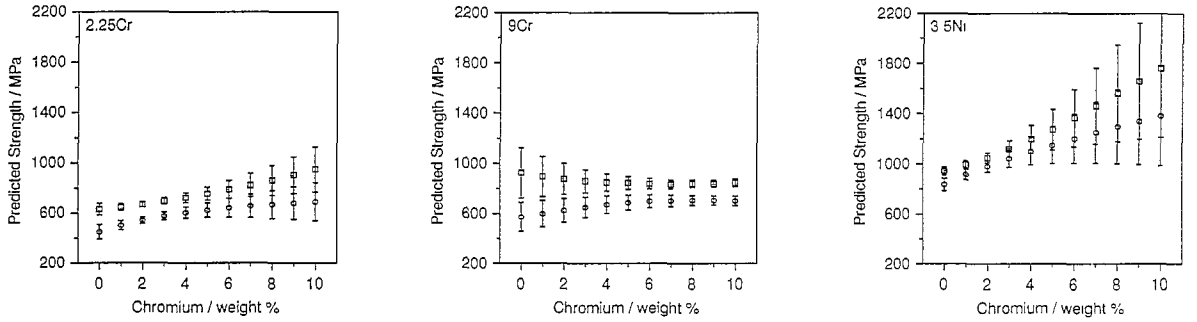


Fig. 16. Predicted variations in yield strength and UTS with chromium variation.

Table 5
MTDATA calculated phases presented at the tempering for the $2\frac{1}{4}$Cr (700°C), 9Cr (760°C) and $3\frac{1}{2}$Ni (700°C) weld deposits with varying chromium concentration

Cr	$2\frac{1}{4}$Cr1Mo					9Cr						$3\frac{1}{2}$Ni			
Wt.%	α	MC	M_7C_3	M_2X	$M_{23}C_6$	α	γ	MC	M_2X	$M_{23}C_6$	M_7C_3	α	γ	M_2X	$M_{23}C_6$
0	99.07	—	0.93	—	—	82.85	16.28	—	0.87	—	—	54.84	44.80	0.36	—
2	98.59	0.03	—	—	1.38	94.85	3.72	—	1.43	—	—	61.80	36.19	0.04	1.97
4	98.60	0.03	—	—	1.37	88.11	10.99	—	—	—	0.90	61.57	36.06	0.04	2.32
6	98.58	—	—	0.05	1.37	97.85	—	0.23	—	1.91	—	59.35	38.23	—	2.42
8	98.54	0.09	—	—	1.37	97.68	—	—	0.40	1.92	—	58.46	39.09	—	2.45
10	98.54	0.10	—	—	1.36	97.84	—	0.23	—	1.93	—	59.09	38.44	—	2.47

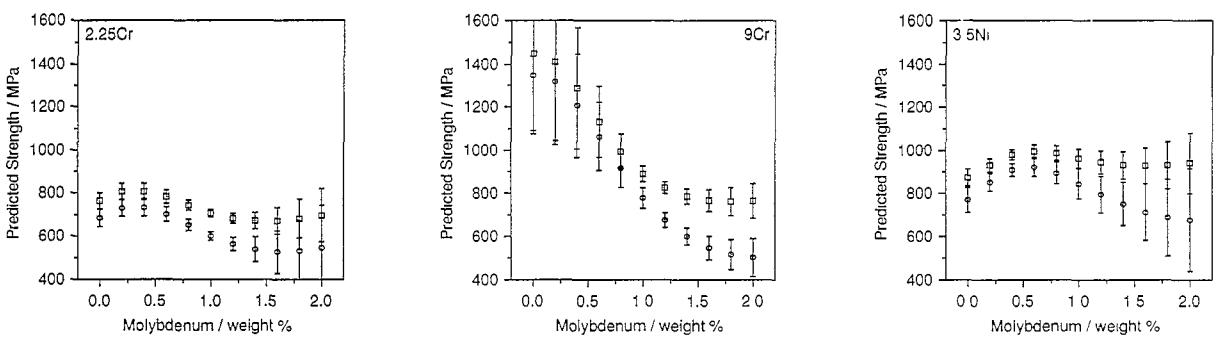


Fig. 17. Predicted variations in yield strength and UTS with molybdenum variation.

increase in the fraction of carbides present in the $3\frac{1}{2}$Ni alloy up to a chromium concentration of about 4 wt.%. This is again consistent with the calculations presented in Fig. 16(c), where the rate of increase in the yield strength is significantly smaller beyond about 4 wt.%, although the calculated values become unreliable at the larger chromium concentrations. For the $2\frac{1}{4}$Cr1Mo alloy, the calculations presented in Fig. 16(a) are at first sight inconsistent with those presented in Table 5, where the carbide fraction does not change greatly as the chromium concentration is increased. However, $M_{23}C_6$ precipitation is known to be notoriously slow [65] in this alloy for reasons described elsewhere [66]. The tempering treatment of 700°C for 3 h is certainly not sufficient to induce $M_{23}C_6$ precipitation [65]. The data in Table 5 are, on the other hand, equilibrium data and therefore do not represent the microstructural state of $2\frac{1}{4}$Cr1Mo after the prescribed tempering heat treatment. This is not the case for the 9Cr alloy, where $M_{23}C_6$ is known to precipitate very rapidly indeed [67]. Unfortunately, the detailed precipitation behaviour in the $3\frac{1}{2}$Ni alloy has never been experimentally characterised, so it is not possible to comment on its microstructural state; the good correlation between the equilibrium carbide fractions and strength calculations therefore is in this case unconfirmed.

The effect of molybdenum is illustrated in Fig. 17. The effects illustrated are experimentally established [60] in the regions where the magnitudes of the error bars are reasonable. The metallurgical reason for this

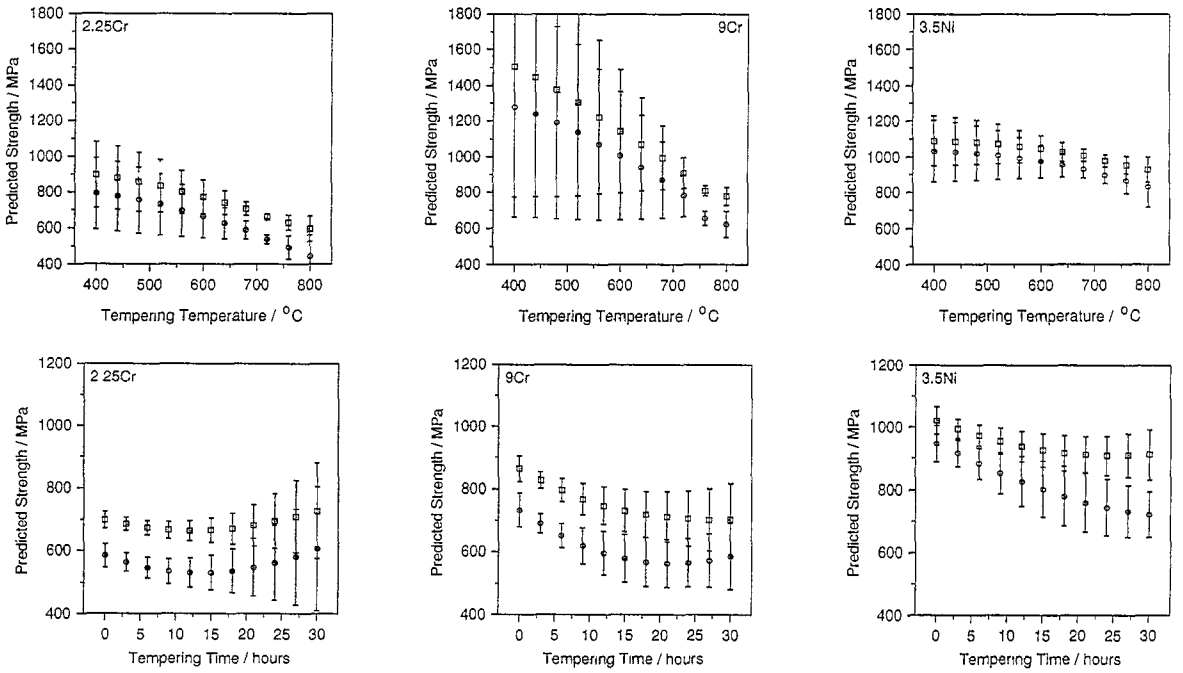


Fig. 18. (a) Predicted varations in yield strength and UTS with tempering (a) temperature variation, (b) time variation.

Table 6
The weights for the highest ranked yield strength model (3 hidden units, random number seed of 35)

−0.171546	−0.00225139	−0.0970205	−0.0300255	−0.0633371	−0.00429571	0.0388401
−0.276171	−0.798268	−0.106679	−0.128003	−0.07281	0.222673	0.324986
0.0424416	−0.0428604	0.171731	0.127469	−0.581542	−0.163026	0.144293
0.016795	−0.141619	0.0484977	0.0786442	0.0212737	0.0325292	0.00343047
−1.16545	−0.0758439	−0.0670774	0.0399578	0.247083	0.338366	0.0107962
−0.0347937	0.199963	−0.0188507	−0.31864	−0.308316	0.255493	−0.00444168
0.025369	−0.039484	−0.136707	0.0123413	0.0539999	−0.184229	0.0590583
−0.00584449	−0.0974913	−0.0842608	−0.0758818	0.147873	0.025533	0.0139334
0.0317911	0.163918	−0.500529	0.0556693	−4.0902	−11.317	8.05724
—	—	—	9.29007	—	—	—

The data are arranged in a continuous horizontal sequence in the following order: $O_1^{(1)}$, $w_{1,1}^{(1)} \cdots w_{1,19}^{(1)}$, $O_2^{(1)}$, $w_{2,1}^{(1)} \cdots w_{2,19}^{(1)}$, $O_3^{(1)}$, $w_{3,1}^{(1)} \cdots w_{3,19}^{(1)}$, $O^{(2)}$, $w_1^{(2)} \cdots w_3^{(2)}$.

Table 7
The weight for the second highest ranked yield strength model (3 hidden units, random number seed of 50)

1.5294	0.130644	0.159576	0.129336	0.0617725	4.25782	1.02219
−0.536674	−1.92137	0.546298	0.714566	−1.66812	−3.26685	−0.0639273
−0.124555	0.447275	1.54761	0.501745	−1.83306	−0.73649	2.53948
0.0837894	0.763305	0.576838	−1.48212	2.05231	0.212393	2.52537
1.99737	1.02329	0.46945	−0.946581	−2.82259	0.219534	−0.119991
1.33101	1.26448	−0.194266	−0.04004	0.0472404	1.44837	−0.0902397
0.630194	0.0726304	−2.43426	4.066	−0.409978	−2.3393	0.838061
0.110044	0.620535	0.824383	−1.13547	0.0976623	0.263277	1.46608
3.11664	0.072773	−0.254449	−0.0203781	−0.745496	0.486221	0.640586
			−0.422727			

The data are arranged in a continuous horizontal sequence in the following order: $O_1^{(1)}$, $w_{1,1}^{(1)} \cdots w_{1,19}^{(1)}$, $O_2^{(1)}$, $w_{2,1}^{(1)} \cdots w_{2,19}^{(1)}$, $O_3^{(1)}$, $w_{3,1}^{(1)} \cdots w_{3,19}^{(1)}$, $O^{(2)}$, $w_1^{(2)} \cdots w_3^{(2)}$.

behaviour is not understood and we were unable to throw light on the behaviour by conducting equilibrium calculations.

Predictions were also made for other carbide forming elements such as vanadium, tungsten and niobium over typical ranges of concentrations; all the calculations

Table 8
The weights for the third highest ranked yield strength model (4 hidden units, random number seed of 35)

1.52771	0.130892	0.158762	0.129669	0.028678	4.28128	1.02017
−0.537522	−1.91699	0.544957	0.710081	−1.65936	−3.27051	−0.0623858
−0.126148	0.448406	1.55414	0.503071	−1.83626	−0.735993	2.54129
0.085005	0.761328	0.57517	−1.47947	2.05066	0.212981	2.52377
1.99755	1.02311	0.466991	−0.927745	−2.83122	0.216813	−0.117682
1.32788	1.26225	−0.194948	−0.0385676	0.0470837	1.45127	−0.0896011
0.628999	0.0700215	−2.43563	4.06835	−0.406778	−2.34629	0.843195
0.111985	0.616167	0.842021	−1.14799	0.0940677	0.267963	1.46457
3.11701	0.0720523	−0.253438	−0.0195078	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	−0.745237	0.486101	0.640934	−0.42255
—	—	—	0	—	—	—

The data are arranged in a continuous horizontal sequence in the following order: $\theta_1^{(1)}$, $w_{1.1}^{(1)} \cdots w_{1\,9}^{(1)}$, $\theta_2^{(1)}$, $w_{2\,1}^{(1)} \cdots w_{2\,9}^{(1)}$, $\theta_3^{(1)}$, $w_{3\,1}^{(1)} \cdots w_{3\,9}^{(1)}$, $\theta_4^{(1)}$, $w_{4.1}^{(1)} \cdots w_{4.9}^{(1)}$, $\theta^{(2)}$, $w_1^{(2)} \cdots w_4^{(2)}$.

Table 9
The weights for the fourth highest ranked yield strength model (4 hidden units, random number seed of 5)

−0.107965	0.0573655	−0.0613495	0.0288485	−0.000134314	0.00305974	0.0836847
0.0406543	−0.516705	−0.0964365	0.100146	−0.000725501	−0.0179112	−0.162536
−0.00557001	−0.0338045	0.0765412	−0.0264282	−0.290584	−0.244032	0.0910335
0.10959	0.0518192	0.00783204	0.00286686	0.000667963	0.107157	0.232409
0.118248	−0.0299825	−0.386075	0.000558905	−0.141753	−0.21818	0.00222818
−0.00391664	0.054927	0.0493295	−0.228877	−0.274673	0.0367457	0.023917
0.0695565	−0.038123	−0.00490637	−0.000783721	0.0255852	−0.00878945	0.241244
0.0320366	−0.294835	−0.000885827	−0.0191008	−0.0335813	0.0193123	0.0221419
−0.0392133	0.0783278	−0.591047	−0.0773039	0.111396	0.0174837	0.0352805
0.0270851	0.00128652	0.00482311	−0.0037441	0.205378	0.340798	0.0452223
−0.129257	−0.000542862	−0.113091	−0.00296279	−0.0146525	0.0160665	0.0117534
−0.0158564	0.621586	0.0665543	1.50679	29.4869	−27.9551	31.4683
—	—	—	33.6464	—	—	—

The data are arranged in a continuous horizontal sequence in the following order: $\theta_1^{(1)}$, $w_{1\,1}^{(1)} \cdots w_{1\,9}^{(1)}$, $\theta_2^{(1)}$, $w_{2.1}^{(1)} \cdots w_{2\,9}^{(1)}$, $\theta_3^{(1)}$, $w_{3\,1}^{(1)} \cdots w_{3\,9}^{(1)}$, $\theta_4^{(1)}$, $w_{4\,1}^{(1)} \cdots w_{4\,9}^{(1)}$, $\theta^{(2)}$, $w_1^{(2)} \cdots w_4^{(2)}$.

Table 10
The weights for the fifth highest ranked yield strength model (2 hidden units, random number seed of 50)

−0.463083	0.0139331	1.38713	0.126085	0.166394	0.224425	0.696231
0.228694	−2.49771	0.0490209	2.80412	−0.727356	−1.13542	−0.692619
−1.35559e−05	0.181066	−0.650902	−0.0167849	−0.0147237	−0.676583	0.115048
0.0127762	−1.36156	−0.00308982	−0.196379	−0.0954772	−0.62722	0.348801
2.97001	0.1547	−3.08881	−0.12653	0.0709729	0.750029	−3.5612e−05
−0.050513	0.6152	−0.0190338	−0.0116097	0.713112	−0.142552	2.72956
—	—	—	2.65188	—	—	—

The data are arranged in a continuous horizontal sequence in the following order: $\theta_1^{(1)}$, $w_{1\,1}^{(1)} \cdots w_{1\,9}^{(1)}$, $\theta_2^{(1)}$, $w_{2\,1}^{(1)} \cdots w_{2\,9}^{(1)}$, $\theta^{(2)}$, $w_1^{(2)} \cdots w_2^{(2)}$.

appeared well-behaved from a metallurgical point of view. For example, the addition of vanadium produced increases in strength due to the precipitation of fine vanadium carbonitrides of the form $V_4(N,C)_3$. These are known to increase the strength of steels [68]. Tungsten, on the other hand, reduced strength, a well known effect originating in the fact that it induces the retention of soft, high-temperature δ-ferrite [69]. Niobium caused insignificant changes in strength for the concentrations studied (less than 1000 ppm wt.), except in the 9Cr alloy were the resulting increase in strength is expected [70].

Finally, it was confirmed that trace impurities such as sulphur and phosphorus have no significant effect on strength. The effect of titanium, boron and copper cannot be discerned reliably in the type of welds studied here.

9. Heat treatment

The effects of tempering temperature and time are as expected, that an increase in either leads to a decrease

Table 11
The weights for the highest ranked ultimate tensile strength model (2 hidden units, random number seed of 35)

−0.545033	−0.0064281	−0.126882	0.0330432	−0.0662099	0.167478	−0.00908487
−0.321078	0.0692095	0.144995	0.0257678	0.000139848	0.000106472	−0.093284
−0.0112513	−0.0188631	0.0317367	−0.107515	0.212019	0.0367566	0.0271643
−0.0661274	0.732212	−0.0151088	−0.0346088	0.117374	0.127426	0.192166
0.0632741	−0.223416	0.0733402	0.256446	0.0661928	8.89202e−05	−0.000106467
−0.148686	0.0250997	0.0861742	0.0664669	−0.0324574	0.180785	−0.0734655
−0.0350267	−0.317535	−0.2994	−0.00220252	0.149785	0.101062	0.218229
−0.0652287	0.0972585	0.184202	6.75444e−05	0.0967146	0.0461672	0.000105812
0.000105008	−0.0536363	0.0387114	0.101838	0.0403639	0.0940998	−0.0696791
−0.126964	−0.0783781	−0.144711	10.5165	7.70683	−7.1856	7.16927

The data are arranged in a continuous horizontal sequence in the following order: $\theta_1^{(1)}$, $w_{1,1}^{(1)} \cdots w_{1,21}^{(1)}$, $\theta_2^{(1)}$, $w_{2,1}^{(1)} \cdots w_{2,21}^{(1)}$, $\theta^{(2)}$, $w_1^{(2)} \cdots w_2^{(2)}$.

Table 12
The weights for the second highest ranked ultimate tensile strength model (5 hidden units, random number seed of 35)

−0.16122	0.0407871	−0.00909901	0.0169438	0.136259	0.0888185	−0.0175177
0.103726	−0.0131323	0.0872891	0.0239744	−0.000826904	−0.00130356	0.215996
0.0496323	0.0167179	0.020286	−0.0135664	0.15469	0.480376	−0.0368901
−0.0747955	−0.172081	0.0897425	−0.0361481	−0.0489297	−0.208196	−0.20474
−0.0579633	−0.0637738	0.0294171	−0.0906785	−0.0646641	0.000636431	0.000926244
0.272689	−0.00147277	0.0826351	−0.0207208	0.0221835	0.0285505	0.626368
0.214549	0.0752131	−1.52502	0.0364549	−0.0374196	−0.0411478	−0.160239
0.0319428	−0.017594	0.270243	−0.0164737	0.0443762	0.125007	0.0004644847
0.000679067	0.537775	−0.0794043	−0.181952	0.0278522	−0.0711348	0.194159
0.12131	−0.197654	0.0873153	−0.00618774	0.0566016	−0.0517417	−0.0930999
−0.549828	−0.349333	−0.0361367	0.083644	0.0601097	−0.173143	−0.0972791
−0.000566209	−0.00780473	0.155943	−0.14056	0.0876987	−0.0118004	−0.0137913
−0.245168	−0.184675	0.0144672	0.207809	−0.317642	−0.0111103	0.0198983
0.0313242	−0.010758	−0.0342228	0.0349095	0.139498	0.0299207	0.0334859
−0.0761596	0.000590194	0.000840134	−0.283084	−0.0157624	0.152502	0.027587
−0.00533164	−0.201626	−0.333052	−0.135756	0.0294736	2.85903	−11.038
—	8.00081	5.57286	−7.80338	7.66909	—	—

The data are arranged in a continuous horizontal sequence in the following order: $\theta_1^{(1)}$, $w_{1,1}^{(1)} \cdots w_{1,21}^{(1)}$, $\theta_2^{(1)}$, $w_{2,1}^{(1)} \cdots w_{2,21}^{(1)}$, $\theta_3^{(1)}$, $w_{3,1}^{(1)} \cdots w_{3,21}^{(1)}$, $\theta_4^{(1)}$, $w_{4,1}^{(1)} \cdots w_{4,21}^{(1)}$, $\theta_5^{(1)}$, $w_{5,1}^{(1)} \cdots w_{5,21}^{(1)}$, $\theta^{(2)}$, $w_1^{(2)} \cdots w_5^{(2)}$.

in the strength (Fig. 18(a) and (b)). The strength does not collapse after prolonged heat treatment because all of the alloys concerned contain strong carbide-forming elements designed to provide secondary hardening and resist tempering.

10. Conclusions

A neural network method based within a Bayesian framework has been used to rationalise an enormous quantity of published experimental data on the yield and tensile strengths of steel weld deposits. It is now possible, therefore, to estimate the strength as a function of the chemical composition, welding conditions and a variety of heat treatment parameters.

The models formulated have been applied towards the understanding of alloys used typically in the construction of power plant. It has been demonstrated that the models not only give good estimates of the strength, but also identify areas which require detailed investigation. The latter is a feature of the Bayesian method which permitted the expression of model-perceived uncertainty as a function of position in the input space.

Although the model has here been applied to power plant alloys, it has much wider applicability since the data used for training purposes included all varieties of welding materials. It has been used successfully on unseen data on welds for submarine applications [71].

Acknowledgements

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Table 13
The weights for the third highest ranked ultimate tensile strength model (5 hidden units, random number seed of 5)

−0.0911319	−0.0467519	−0.185872	0.0542093	0.139544	0.112131	−0.00884624
−0.159532	0.0762567	−0.0752384	0.0329026	−6.76202e−05	−0.000832395	0.10332
0.0176329	0.457008	−0.0952556	0.307908	0.0127599	0.00153671	−0.0344478
0.0353183	−0.0293909	0.0921828	0.23866	−0.157204	−0.254759	−0.301421
−0.132347	0.373071	0.0449085	−0.0840789	−0.0743927	0.000103465	0.00130424
0.0281757	−0.0607581	0.0845989	0.030228	−0.155432	−0.0989584	0.407892
0.121197	0.072573	−0.0334251	0.117165	0.091334	0.0607349	−0.329338
−0.150816	−0.0344293	0.0254348	0.0643187	0.0170009	0.0785463	−0.000140272
−0.0022139	−0.140086	0.0239122	0.247511	0.0303347	−0.204584	−0.128115
0.108966	−0.0090146	−0.00418695	−0.0371955	−0.0101928	0.270777	−0.147764
0.10708	−0.10376	−0.0364737	0.156751	−0.0886395	0.031076	−0.118679
−0.000104144	−0.00229921	0.0265603	0.00725928	−0.340225	0.0552878	−0.179458
0.118286	0.295933	0.166681	−0.13071	−0.276006	−0.0461078	−0.00489961
0.136872	0.263665	0.188504	0.0881901	−0.397621	−0.0242848	0.111534
0.0567841	0.000213839	0.00178291	−0.0580242	0.13678	0.2538	−0.0149016
0.0369014	0.18085	0.00767431	0.0329047	−0.177406	1.52726	−8.9659
—	9.70672	−8.12352	−10.5055	9.05696	—	—

The data are arranged in a continuous horizontal sequence in the following order: $\theta_1^{(1)}$, $w_{1\,1}^{(1)} \cdots w_{1\,21}^{(1)}$, $\theta_2^{(1)}$, $w_{2\,1}^{(1)} \cdots w_{2\,21}^{(1)}$, $\theta_3^{(1)}$, $w_{3\,1}^{(1)} \cdots w_{3\,21}^{(1)}$, $\theta_4^{(1)}$, $w_{4.1}^{(1)} \cdots w_{4\,21}^{(1)}$, $\theta_5^{(1)}$, $w_{5.1}^{(1)} \cdots w_{5.21}^{(1)}$, $\theta^{(2)}$, $w_1^{(2)} \cdots w_5^{(2)}$.

Table 14
The weights for the fourth highest ranked ultimate tensile strength model 92 hidden units, random number seed of 35)

−0.341103	0.000208627	−0.0366557	−0.0943532	−0.0174077	−0.00562245	−0.0421953
0.20631	−0.0645495	−0.180316	−0.0477751	−0.0148117	0.00120155	0.0953614
−0.0794944	−0.0461336	−0.0320866	−0.00411833	−0.072802	0.0127794	−0.00137493
0.233802	−0.626333	0.00772527	0.0703737	0.140811	0.042378	−0.0220453
0.071239	−0.280358	0.105118	0.256957	0.0726967	0.0224132	0.000835579
−0.14672	0.101595	0.060798	0.0549941	0.00194871	0.0995868	−0.0181281
—	−0.000927529	−0.236674	9.49514	9.69405	11.3157	—

The data are arranged in a continuous horizontal sequence in the following order: $\theta_1^{(1)}$, $w_{1\,1}^{(1)} \cdots w_{1.21}^{(1)}$, $\theta_2^{(1)}$, $w_{2\,1}^{(1)} \cdots w_{2.21}^{(1)}$, $\theta^{(2)}$, $w_1^{(2)} \cdots w_2^{(2)}$.

Appendix A

The weights for all components of the yield strength and UTS committee models which have been trained using the *entire* dataset (Tables 6–14).

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